

Panic Attack Analysis Dashboard — Project Report

1. Introduction

The **Panic Attack Analysis Dashboard** is a data-driven project designed to analyze panic attacks using anonymized patient data. Leveraging over **1,200 patient records**, the dashboard helps identify patterns in symptoms, triggers, demographics, and lifestyle factors through interactive visualizations in **Power BI**.

This project aims to assist researchers, healthcare professionals, and mental health advocates in understanding panic attack dynamics to improve patient care.

2. Background: Understanding Panic Attacks

- Panic attacks are **sudden and intense episodes of fear or anxiety**.
 - Symptoms commonly include:
 - Rapid heart rate
 - Difficulty breathing
 - Dizziness
 - Sweating
 - Chest pain
 - Trembling
 - These episodes can occur **without warning** or any obvious trigger.
 - Frequent panic attacks can lead to **panic disorder**, characterized by persistent fear of future attacks.
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3. Data and Technology

- Dataset: `panic_attack_dataset.csv` containing over 1,200 anonymized patient entries.
 - Data Warehouse: **Snowflake** used for scalable data storage and ETL processes.
 - Visualization: **Power BI** dashboard with live connection to Snowflake for real-time filtering and dynamic reports.
 - Key data fields include patient demographics, panic attack frequency, symptoms, triggers, lifestyle habits, and medical history.
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4. Data Pipeline and Transformation

- **ETL Process:**
Raw CSV data was cleaned (handling nulls, duplicates) and uploaded to Snowflake for governance and scalability.
- **Key Enhancements:**
 - Panic attacks categorized into **High (7–9)**, **Medium (4–6)**, and **Low (0–3)** frequency bins for better analysis.

- Heart rate values corrected (adjusted down by 1 bpm) to fix a data glitch, resulting in realistic averages (120–140 bpm).

5. Key Findings

Symptoms Prevalence

Symptom	Percentage of Patients
Dizziness	52%
Trembling	48%
Chest Pain	40%
Shortness of Breath	38%
Sweating	30%

Notably, dizziness and shortness of breath commonly co-occur in PTSD cases.

Top Triggers by Frequency

- Unknown, Caffeine, Phobia (~17% each)
- PTSD (17%)
- Social Anxiety, Stress (16% each)

Demographics & Habits

- Females report **higher panic attack frequencies**.
- Smokers experience **50% more sweating** during attacks.
- Medical history prevalence:
 - Anxiety (45%)
 - PTSD (30%)
 - Depression (20%)
- Patients undergoing therapy show a **15% lower severity score**.

Lifestyle Correlations

- Less than 6 hours of sleep correlates with a **20% increase** in attack frequency.
- Poor caffeine management and low exercise frequency are linked to increased panic risk.

6. Actionable Insights

- Patients with **high-frequency attacks (7 or more)** are **3 times more likely** to have PTSD.
- Lifestyle changes such as **reducing caffeine intake** and **improving sleep quality** could reduce panic triggers by **25–30%**.

7. Technologies Utilized

- **Power BI** — Interactive, dynamic dashboards.
 - **Snowflake** — Cloud data warehousing and ETL.
 - **DAX** — Custom calculations for data correction and categorization.
 - **CSV** — Raw patient data format.
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8. Conclusion

The Panic Attack Analysis Dashboard provides a comprehensive tool for understanding panic attack trends and potential interventions. By combining data cleaning, categorization, and powerful visual analytics, this project highlights critical correlations between symptoms, triggers, lifestyle, and demographics that can guide better mental health support and research.